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GATEMATE - ADVANCED ASSEMBLY GUIDE
Modular 3D-Printed Connector System for FPV Gates

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Product Status: Prototyping Phase 稱
Target Hardware: Standard 20mm PVC Electrical Conduits (Baumarkt)
Document Version: 1.1 (Beta)

1. PRODUCT OVERVIEW

GateMate is designed to make DIY FPV drone racing gates affordable, lightweight, and highly portable. By replacing expensive retail corner brackets with custom 3D-printed connectors, you can build infinite track layouts using cheap 20mm PVC pipes from your local hardware store.

Whether you are blasting through your basement with a 65mm Tiny Whoop or tearing up a field with a 6S 5-inch freestyle quad, GateMate connectors are optimized to withstand heavy crashes without breaking.

2. ADVANCED 3D PRINTING SETTINGS (CRASH-RESISTANT)

To ensure your gates don't shatter upon high-velocity impacts, do NOT use standard slicing profiles. Follow these strength-optimized settings:

- * Material Recommendations:
 - PETG: Excellent outdoor durability, high flexibility (bends instead of snapping upon impact). Highly recommended for 5-inch gates.
 - PLA: Easy to print. Great for Whoop tracks Also works good for 5-inch.
- * Slicer Configuration:
 - Perimeters (Walls): Minimum 4-5 layers. (More walls = more strength!)
 - Top/Bottom Layers: 4 layers minimum.
 - Infill Percentage: 30% to 45%.
 - Infill Pattern: Gyroid.
 - Print Temperature: Print at the higher end of your filament's spec sheet to maximize layer adhesion.
 - no supports: the prints are optimized so modern printers can print them without supports.

3. REQUIRED HARDWARE & MATERIALS

To complete a standard square or arch gate, you will need:

- * PVC Pipes: Standard 20mm outer diameter (OD) plastic conduits.
- * Tools: A basic hacksaw or a pipe cutter to cut the PVC tubes.
- * Sandpaper: To smooth out the cut edges of the PVC pipes for an easier fit into the 3D prints.

4. ASSEMBLY & FITMENT TIPS

- * Friction Fit: The connectors are engineered with tight tolerances so the pipes stay in place without glue.
- * Easy Disassembly: If the fit is too tight due to printer over-extrusion, slightly sand down the outside of the PVC pipe ends or use a heat gun to gently warm the 3D print before inserting.
- * Portability: Do not glue the joints! Leave them as a friction fit so you can pack down an entire 5-inch gate setup into a backpack.

5. ROADMAP & UPCOMING FEATURES

The GateMate ecosystem is constantly evolving during this prototyping phase. Future releases will include:

- 90-Degree Corner Matrix Connectors
- 45-Degree Sharp Angle Brackets (for aggressive racing lines)
- Weighted T-Base Mounts for asphalt or indoor sports halls
- Official STL listings on Cults3D and Printables Store

Stay tuned! The final PDF manual will feature full 3D technical drawings, exact cutting blueprints, and official race layouts.

Follow the development logs on Instagram and YouTube!

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